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Developing cartridge provided with cover

Abstract

A process cartridge including: a photosensitive member cartridge including a photosensitive member; and a developing cartridge including a developer carrier. The developing cartridge further includes: an inputting portion; a first cover; and a second cover. The first cover has a first engagement portion configured to move the developer carrier away from the photosensitive member. The first engagement portion is positioned between an axis of the developer carrier extending in an axial direction and an axis of the inputting portion extending in the axial direction, when projected in the axial direction. The second cover has a second engagement portion configured to move the developer carrier away from the photosensitive member. The second engagement portion is positioned between the axis of the developer carrier and the axis of the inputting portion, when projected in the axial direction.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation application of U.S. patent application Ser. No. 17/475,485, filed Sep. 15, 2021, now U.S. Pat. No. 11,733,646 issued Aug. 22, 2023, which is a continuation application U.S. patent application Ser. No. 17/155,396, filed Jan. 22, 2021, now U.S. Pat. No. 11,669,043, issued Jun. 6, 2023, which is a continuation application of U.S. patent application Ser. No. 16/829,909, filed Mar. 25, 2020, now U.S. Pat. No. 10,901,361, issued Jan. 26, 2021, which is a continuation application of U.S. patent application Ser. No. 16/408,999, filed May 10, 2019, now U.S. Pat. No. 10,620,580, issued Apr. 14, 2020, which is a continuation application of U.S. patent application Ser. No. 16/032,135, filed Jul. 11, 2018, now U.S. Pat. No. 10,303,112, issued May 28, 2019, which is a continuation of U.S. patent application Ser. No. 15/866,939, filed Jan. 10, 2018, now U.S. Pat. No. 10,048,645, issued Aug. 14, 2018, which is a continuation of U.S. patent application Ser. No. 15/351,827, filed Nov. 15, 2016, now U.S. Pat. No. 9,891,581, issued Feb. 13, 2018, which is a continuation of U.S. patent application Ser. No. 15/078,362, filed Mar. 23, 2016, now U.S. Pat. No. 9,529,302, issued Dec. 27, 2016, which is a continuation of U.S. patent application Ser. No. 13/599,335, filed Aug. 30, 2012, now U.S. Pat. No. 9,323,214, issued Apr. 26, 2016, which claims priority from Japanese Patent Application No. 2011-190037 filed Aug. 31, 2011. The contents of the above noted applications are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(1) The present invention relates to a process cartridge and a developing cartridge, detachably mountable in an electrophotographic type image forming device.

BACKGROUND

(2) Known is an electrophotographic type printer provided with a process cartridge. The process cartridge includes a drum cartridge having a photosensitive drum, and a developing cartridge having a developing roller for supplying toner to the photosensitive drum. The process cartridge is detachably mountable in the printer.

(3) Such a printer has a known configuration such that the developing roller of the developing cartridge contacts the photosensitive drum to supply the toner to the photosensitive drum when an image is formed, and is separated from the photosensitive drum not to supply the toner to the photosensitive drum when an image is not formed.

(4) For example, the developing cartridge has right and left side walls on which engagement protrusions are respectively provided. The drum cartridge has right and left side walls on which front nipping levers and rear nipping levers are respectively provided. Each engagement protrusion is nipped by the front nipping lever and the rear nipping lever, so that the developing cartridge is brought into contact with and separated from the photosensitive drum.

(5) With this configuration, the rear nipping levers respectively press the engagement protrusions of

the developing cartridge toward the photosensitive drum. Thus, the developing cartridge normally contacts the photosensitive drum. Further, to separate the developing cartridge from the photosensitive drum, the front nipping levers respectively press the engagement protrusions of the developing cartridge forward against the pressing force from the rear nipping levers.

SUMMARY

(6) In the above-described configuration, it is required to provide the engagement protrusions on both of the right and left side walls of the developing cartridge.

(7) In general, there are many parts and components on the right and left side walls of the developing cartridge for receiving a drive force and an electric power supplied from a main casing side, other than the engagement protrusions.

(8) For this reason, when downsizing of the developing cartridge is attempted, it is difficult to secure a space for installing the engagement protrusions on the right and left side walls of the developing cartridge. This may decrease a degree of freedom in designing the developing cartridge.

(9) In view of the foregoing, it is an object of the present invention to provide a process cartridge and a developing cartridge, capable of enhancing a degree of freedom in design of the developing cartridge while downsizing the developing cartridge.

(10) In order to attain the above and other objects, the present invention provides a process cartridge including: a photosensitive member cartridge; and a developing cartridge. The photosensitive member cartridge includes a photosensitive member. The developing cartridge includes: a developing cartridge frame; a developer carrier; a first gear; a second gear; an inputting portion; a first cover; and a second cover. The developing cartridge frame defines an internal space for accommodating a developing agent therein. The developer carrier is rotatably supported in the developing cartridge frame and arranged in confrontation with the photosensitive member to supply the developing agent to the photosensitive member. The developer carrier has an axis extending in an axial direction. The developing cartridge frame has a first side and a second side opposite to the first side in the axial direction. The first gear is disposed at the first side. The second gear is disposed at the second side. The inputting portion has an axis parallel to the axial direction and is rotatable about the axis of the inputting portion in response to an external drive force to transmit the external drive force to the first gear. The first cover covers at least a part of the first gear and has a first engagement portion configured to move the developer carrier away from the photosensitive member. The first engagement portion is positioned between the axis of the developer carrier and the axis of the inputting portion when projected in the axial direction. The second cover covers at least a part of the second gear and has a second engagement portion configured to move the developer carrier away from the photosensitive member. The second engagement portion is positioned between the axis of the developer carrier and the axis of the inputting portion when projected in the axial direction.

(11) According to another aspect, the present invention provides a developing cartridge including: a developing cartridge frame; a developer carrier; a first gear; a second gear; an inputting portion; a first cover; and a second cover. The developing cartridge frame defines an internal space for accommodating a developing agent therein. The developer carrier is configured to rotate and has an axis extending in an axial direction. The developing cartridge frame has a first side and a second side opposite to the first side in the axial direction. The first gear is disposed at the first side. The second gear is disposed at the second side. The inputting portion has an axis parallel to the axial direction and is rotatable about the axis of the inputting portion in response to an external drive force to transmit the external drive force to the first gear. The first cover covers at least a part of the first gear and has a first engagement portion configured to move the developer carrier in a perpendicular direction perpendicular to the axial direction. The first engagement portion is positioned between the axis of the developer carrier and the axis of the inputting portion when projected in the axial direction. The second cover covers at least a part of the second gear and has a second engagement portion configured to move the developer carrier in the perpendicular direction.

The second engagement portion is positioned between the axis of the developer carrier and the axis of the inputting portion when projected in the axial direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The particular features and advantages of the invention as well as other objects will become apparent from the following description taken in connection with the accompanying drawings, in which:
- (2) FIG. 1 is a cross-sectional view of a printer in which a process cartridge according to a first embodiment of the present invention is provided;
- (3) FIG. 2 is a perspective view of a drum cartridge shown in FIG. 1, as viewed from an upper rear side thereof;
- (4) FIG. 3 is a perspective view of a developing cartridge shown in FIG. 1, as viewed from an upper left side thereof;
- (5) FIG. 4 is a perspective view of the developing cartridge shown in FIG. 1, as viewed from an upper right side thereof;
- (6) FIG. 5 is an explanatory diagram illustrating attachment of the developing cartridge relative to the drum cartridge;
- (7) FIGS. 6A and 6B are explanatory diagrams illustrating guiding of the developing cartridge by a lock lever, in which FIG. 6A is an enlarged view of a left portion of the developing cartridge in the process of being guided by the lock lever, and FIG. 6B is an enlarged view of the left portion of the developing cartridge after having been guided by the lock lever;
- (8) FIG. 7 is a left side view of the process cartridge in such a state that the process cartridge is mounted in the printer shown in FIG. 1, in which a developing roller and a photosensitive drum are in contact with each other (contact state);
- (9) FIG. 8 is a right side view of the process cartridge in such a state that the process cartridge is mounted in the printer shown in FIG. 1, in which the developing roller and the photosensitive drum are in contact with each other (contact state);
- (10) FIG. 9 is a left side view of the process cartridge in such a state that the process cartridge is mounted in the printer shown in FIG. 1, in which the developing roller and the photosensitive drum are spaced apart from each other (separation state);
- (11) FIG. 10 is a perspective view of process cartridge according to a second embodiment of the present invention, as viewed from an upper left side thereof;
- (12) FIG. 11 is an explanatory diagram illustrating pressure from the developing cartridge relative to the photosensitive drum in the process cartridge shown in FIG. 10;
- (13) FIG. 12 is a perspective view of a developing cartridge according to a third embodiment of the present invention, as viewed from an upper left side thereof;
- (14) FIG. 13 is an explanatory diagram illustrating attachment of the developing cartridge shown in FIG. 12 to a drum cartridge, in which a rear end portion of the developing cartridge is not sufficiently inserted into a cartridge mounting portion of the drum cartridge, and an operation guide of the developing cartridge is in contact with a front wall of the drum cartridge from a top side thereof;
- (15) FIG. 14 is an explanatory diagram illustrating the attachment of the developing cartridge shown in FIG. 12 to the drum cartridge, following FIG. 13, in which the rear end portion of the developing cartridge is sufficiently mounted in the cartridge mounting portion of the drum cartridge, and the operation guide is positioned rearward of the front wall of the drum cartridge;
- (16) FIG. 15 is an explanatory diagram illustrating the attachment of the developing cartridge shown in FIG. 12 to the drum cartridge, following FIG. 14, in which attachment of the developing

cartridge to the drum cartridge has been completed; and

(17) FIG. 16 is an explanatory diagram illustrating detachment of the developing cartridge shown in FIG. 12 from the drum cartridge.

DETAILED DESCRIPTION

(18) A process cartridge and a developing cartridge according to a first embodiment of the present invention, detachably mountable in an image forming device, will be described while referring to FIGS. 1 through 9 wherein like parts and components are designated by the same reference numerals to avoid duplicating description.

(19) 1. Overall Structure of Color Printer

(20) As shown in FIG. 1, the image forming device is a horizontal direct tandem type color printer 1.

(21) The terms “upward”, “downward”, “upper”, “lower”, “above”, “below”, “beneath”, “right”, “left”, “front”, “rear” and the like will be used throughout the description assuming that the printer 1 is disposed in an orientation in which it is intended to be used. In the following description, the right side in FIG. 1 will be referred to as the front side of the printer 1, and the left side in FIG. 1 will be referred to as the rear side of the printer 1. Top, bottom, left, and right sides of the printer 1 in the following description will be based on the reference point of a user viewing the printer 1 from the front side. The near side in FIG. 1 will be referred to as the left side of the printer 1, and the far side in FIG. 1 will be referred to as the right side of the printer 1.

(22) The printer 1 includes a main casing 2 formed in a generally box-shape. The main casing 2 has a top portion at which a top cover 6 is provided. The top cover 6 is pivotally movable about its rear portion between a closed position for closing an opening 5 formed in the main casing 2 and an open position for opening the opening 5. The printer 1 further includes four process cartridges 11 corresponding to each color.

(23) Four of the process cartridges 11 are detachably mounted in the main casing 2, and juxtaposed with each other in a frontward/rearward direction with a space between neighboring process cartridge 11. The process cartridges 11 are disposed at positions above a sheet supply unit 3. Specifically, the process cartridges 11 include a black process cartridge 11K, a yellow process cartridge 11Y, a magenta process cartridge 11M, and a cyan process cartridge 11C arranged in this order from front to rear.

(24) Further, each process cartridge 11 includes a drum cartridge 24 and a developing cartridge 25 detachably mountable in the drum cartridge 24.

(25) The drum cartridge 24 includes a photosensitive drum 15.

(26) The photosensitive drum 15 is formed in a cylindrical shape that is elongated in a rightward/leftward direction (lateral direction). The photosensitive drum 15 is rotatably supported in the drum cartridge 24.

(27) The developing cartridge 25 includes a developing roller 16.

(28) The developing roller 16 includes a developing roller shaft 30 made of metal and extending in the rightward/leftward direction. That is, the developing roller 16 is oriented with its axis in the rightward/leftward direction. The rightward/leftward direction corresponds to an axial direction. The developing roller 16 is rotatably supported to the developing cartridge 25 so that a rear edge of the developing roller 16 is exposed through a rear edge of the developing cartridge 25 and contacts the corresponding photosensitive drum 15 from an upper front side thereof. That is, the developing roller 16 is in confrontation with the photosensitive drum 15. The developing roller 16 is rotatable about an axis A1 (FIGS. 7 and 8) of the developing roller shaft 30 extending in the rightward/leftward direction, that is, in the axial direction.

(29) Further, the developing cartridge 25 includes a supply roller 27 for supplying toner to the developing roller 16, and a thickness-regulating blade 28 for regulating the thickness of the toner supplied to the developing roller 16. Toner is accommodated in a space defined above the supply roller 27 and the thickness-regulating blade 28. That is, toner is accommodated in an internal space

defined by a cartridge frame **51** (described later) of the developing cartridge **25**.

(30) The toner accommodated in the developing cartridge **25** is supplied onto the supply roller **27**, which in turn supplies the toner to the developing roller **16**. The toner is positively turbocharged between the supply roller **27** and the developing roller **16**. A uniform thin layer of toner is carried on a surface of the developing roller **16**.

(31) In the meantime, a Scorotron charger **26** applies a uniform charge of positive polarity to a surface of the corresponding photosensitive drum **15**. Subsequently, an LED unit **12** exposes the surface of the corresponding photosensitive drum **15** to light based on prescribed image data. An electrostatic latent image corresponding to the image data is formed on the surface of the photosensitive drum **15**. The toner carried on the surface of the developing roller **16** is supplied to the electrostatic latent image formed on the surface of the photosensitive drum thereby forming a toner image (developing agent image) on the surface of the photosensitive drum **15**.

(32) A sheet supply tray **7** is disposed at a bottom portion of the main casing **2** and accommodates sheets of paper **S** therein. Each sheet **S** is conveyed upward and then rearward along a U-shaped path by a pickup roller **8**, a sheet supply roller **9**, and a pair of registration rollers **10**, and further conveyed toward a position between the photosensitive drums **15** and a conveying belt **19** at a prescribed timing. The conveying belt **19** conveys the sheet **S** rearward so that the sheet **S** passes sequentially through each position between the photosensitive drums and corresponding transfer rollers **20**. At this time, toner images in each color carried on the respective photosensitive drums **15** are sequentially transferred onto the sheet **S** to form a color image.

(33) As the sheet **S** passes between a heating roller **21** and a pressure roller **22**, the color image is thermally fixed onto the sheet **S** by heat and pressure.

(34) After the color image has been fixed onto the sheet **S**, the sheet **S** is conveyed upward and then frontward along a U-shaped path to be discharged onto a discharge tray **23** provided at the top cover **6**.

(35) 2. Process Cartridge

(36) (1) Drum Cartridge

(37) (1-1) Drum Frame

(38) Note that directions related to the drum cartridge **24** in the following description will be referred based on its position when the drum cartridge **24** is disposed at a horizontal plane in an orientation such that a bottom wall **35** of the drum cartridge **24** is positioned at a bottom side (FIG. 2), unless otherwise specified. A side of the drum cartridge **24** at which the photosensitive drum **15** is disposed will be referred to as a rear side.

(39) As shown in FIG. 2, the drum cartridge **24** includes a drum frame **31**. The drum frame **31** is formed in a generally rectangular frame-like shape with a bottom wall.

(40) The drum frame **31** has a right and left pair of side walls **32**, a front wall **33**, a bottom wall **35**, and a top wall **34**. Hereinafter, the side wall **32** on the right side will be referred to as the right side wall **32R**, and the side wall **32** on the left side will be referred to as a left side wall **32L** when it is necessary to distinguish between the two.

(41) Each of the side walls **32R**, **32L** is formed in a generally rectangular shape in a side view and elongated in the frontward/rearward direction (specifically, in a direction from an upper front side to a lower rear side of the drum frame **31**).

(42) The front wall **33** bridges a front edge of the right side wall **32R** and a front edge of the left side wall **32L**. The front wall **33** is formed in a generally flat plate shape that is elongated in the rightward/leftward direction. The front wall **33** is provided with two pressure member retaining portions **48**. Within each of the pressure member retaining portions **48**, a pressure member **49** is retained.

(43) One of the pressure member retaining portions **48** is disposed at a right end portion of the front wall **33**, and remaining one of the pressure member retaining portions **48** is disposed at a left end portion of the front wall **33**. Each of the pressure member retaining portions **48** is formed in a

generally rectangular shape in a front view. More specifically, each of the pressure member retaining portions **48** is depressed frontward from a rear surface of the front wall **33**.

(44) Each pressure member **49** is formed in a generally square pillar shape in a front view. Each pressure member **49** is urged by an urging member **50** (FIG. 7) so as to normally protrude rearward (diagonally below and rearward in FIG. 7) from the corresponding pressure member retaining portion **48**.

(45) The bottom wall **35** bridges a bottom edge of the right side wall **32R** and a bottom edge of the left side wall **32L**. The bottom wall **35** is connected to a bottom edge of the front wall **33**. The bottom wall **35** is formed in a generally flat plate shape that is elongated in the frontward/rearward direction and in the rightward/leftward direction.

(46) The top wall **34** bridges an upper edge of a rear portion of the right side wall **32R** and an upper edge of a rear portion of the left side wall **32L**. The top wall **34** is formed in a generally flat plate shape that is elongated in the rightward/leftward direction. The top wall **34** is disposed so as to cover the photosensitive drum **15** from a top side thereof. Further, the top wall **34** supports the Scorotron charger **26**.

(47) Within the drum frame **31**, a developing cartridge mounting portion **36** is defined by a front half portion of the bottom wall **35**, the pair of side walls **32** and the front wall **33** corresponding to the front half portion of the bottom wall **35**, and a front edge of the top wall **34**. The developing cartridge mounting portion **36** is adapted to accommodate the developing cartridge **25** therein.

(48) (1-2) Lock Lever

(49) A lock lever **38** as is provided at a front end portion of the developing cartridge mounting portion **36** of the drum cartridge **24** at a position rightward (laterally inward) of the left side wall **32L**. The lock lever **38** is adapted to maintain the developing cartridge **25** in a mounted state.

(50) The lock lever **38** is integrally provided with a pivot shaft **40**, an operation portion **41** extending upward from the pivot shaft **40**, and a lift portion **42** extending diagonally below and rearward from the pivot shaft **40**.

(51) The pivot shaft **40** is formed in a generally cylindrical shape extending in the rightward/leftward direction.

(52) The operation portion **41** is formed in a generally lever shape extending upward from the pivot shaft **40**. The operation portion **41** protrudes upward than an upper edge of the left side wall **32L**. The operation portion **41** has an upper portion at which a restricting portion **43** is provided. The restricting portion **43** is formed in a generally flat plate shape that is elongated in the rightward/leftward direction. The restricting portion **43** has a right edge that protrudes rightward than a right edge of the pivot shaft **40**. The right edge of the restricting portion **43** thus protrudes into an attachment and detachment path of the developing cartridge **25** relative to the drum cartridge **24** (FIG. 6A).

(53) The lift portion **42** is formed in a generally lever shape extending diagonally below and rearward from the pivot shaft **40**.

(54) The lock lever **38** is pivotally movably supported to the left side wall **32L** by a left end portion of the pivot shaft **40**.

(55) The lock lever **38** is pivotally movable about an axis of the pivot shaft **40** between a lock position (FIG. 2) in which the operation portion **41** upstands and an unlock position (not shown) in which the operation portion **41** is inclined.

(56) The lock lever **38** is urged by an urging member (not shown) in a clockwise direction as viewed from a right side, so that the lock lever **38** is normally positioned at the lock position.

(57) (1-3) Separating Member

(58) As shown in FIGS. 2 and 5, a right and left pair of separating members **44** is provided at a rear end portion of the developing cartridge mounting portion **36** of the drum cartridge **24**. More specifically, the right separating member **44** is disposed at an outer surface of the right side wall **32R**, and the left separating member **44** is disposed at an outer surface of the left side wall **32L**.

(59) Each separating member **44** is integrally provided with a pivot shaft **45**, a contacted portion **46**, and a pressing portion **47**.

(60) The pivot shaft **45** is formed in a generally cylindrical shape extending in the rightward/leftward direction.

(61) The contacted portion **46** is formed in a generally flat plate shape extending downward from a lower portion of the pivot shaft **45**. The contacted portion **46** has a lower portion bending forward.

(62) The pressing portion **47** is formed in a generally flat plate shape extending upward from an upper portion of the pivot shaft **45**. The pressing portion **47** has an upper portion bending forward.

(63) Each separating member **44** is supported to the corresponding side wall **32** and pivotally movable about an axis of the pivot shaft **45**.

(64) Each separating member **44** is pivotally movable about the axis of the pivot shaft **45** between a pressure position in which the pressing portion **47** protrudes frontward (diagonally above and frontward in FIG. **9**) and a pressure release position in which the pressing portion **47** is retracted rearward (diagonally below and rearward in FIG. **7**).

(65) (2) Developing Cartridge

(66) Note that, unless otherwise specified, directions related to the developing cartridge **25** in the following description will be referred based on its position when the developing cartridge is disposed at a horizontal plane in an orientation such that a bottom wall **58** of the developing cartridge **25** is positioned at a bottom side thereof (FIG. **3**). Further, a side of the developing cartridge **25** at which the developing roller **16** is positioned will be referred to as a rear side, and a side of the developing cartridge **25** at which the thickness-regulating blade **28** is positioned will be referred to as a top side.

(67) As shown in FIGS. **3** and **4**, the developing cartridge **25** includes the cartridge frame **51**, a drive unit **52** provided at a position leftward of the cartridge frame **51**, and a detection unit **53** provided at a position rightward of the cartridge frame **51**.

(68) (2-1) Cartridge Frame

(69) The cartridge frame **51** is formed in a generally box shape that is elongated in the rightward/leftward direction. The developing roller **16** is rotatably supported to the cartridge frame **51**. The cartridge frame **51** is provided with a first frame **54** constituting a lower portion of the cartridge frame **51**, and a second frame **55** constituting an upper portion of the cartridge frame **51**.

(70) The first frame **54** is formed in a generally box shape with top and rear openings. The first frame **54** is integrally provided with a right and left pair of side walls **56**, a front wall **57**, and a bottom wall **58**.

(71) Hereinafter, the side wall **56** on the right side will be referred to as the right side wall **56R**, and the side wall **56** on the left side will be referred to as a left side wall **56L** when it is necessary to distinguish between the two.

(72) Each side wall **56** is formed in a rectangular shape in a side view that is elongated in a vertical direction and in the frontward/rearward direction. The right side wall **56R** and the left side wall **56L** is arranged in confrontation with and spaced apart from each other in the rightward/leftward direction.

(73) The front wall **57** is elongated in the rightward/leftward direction, and bridges a front edge of the right side wall **56R** and a front edge of the left side wall **56L**.

(74) The bottom wall **58** is elongated in the rightward/leftward direction. The bottom wall **58** is connected to a lower edge of the front wall **57**, and bridges a lower edge of the right side wall **56R** and a lower edge of the left side wall **56L**.

(75) The second frame **55** is formed in a generally rectangular flat plate shape in a plan view, and connected to front portions of the right and left side walls **56** and an upper edge of the front wall **57**. The second frame **55** has a rear portion at which the thickness-regulating blade **28** is disposed such that the thickness-regulating blade **28** contacts the developing roller **16** from a top side thereof.

(76) (2-2) Drive Unit

(77) As shown in FIGS. 3 and 5, the drive unit 52 includes a developing coupling 61 and a drive unit side gear cover 62.

(78) The developing coupling 61 is rotatably supported to the left side wall 56L. The developing coupling 61 is integrally provided with a coupling portion 60 and a gear portion 64. The developing coupling 61 is rotatable about an axis A2 (FIGS. 7 and 8) of the coupling portion 60.

(79) The coupling portion 60 is formed in a generally cylindrical shape extending in the rightward/leftward direction. The coupling portion 60 has a left end wall formed with a recessed connection portion 63.

(80) The recessed connection portion 63 is depressed rightward from the left end wall of the coupling portion 60 and elongated in the radial direction of the coupling portion 60. The recessed connection portion 63 has a generally elongated shape in a side view with a center portion in the radial direction having a narrow width. Incidentally, the main casing 2 is provided with a main casing coupling (not shown), and a leading end of the main casing coupling is non-rotatably inserted into the recessed connection portion 63 when the developing cartridge 25 is mounted in the main casing 2. A drive force generated on the main casing 2 side is inputted into the developing coupling 61 through the main casing coupling (not shown).

(81) The gear portion 64 is disposed at a right end portion of the coupling portion 60 across the entire outer peripheral surface thereof. The gear portion 64 is adapted to transmit a drive force to the developing roller 16 and the supply roller 27 via a gear train (not shown).

(82) The drive unit side gear cover 62 is formed in a generally square pillar shape whose left end is closed, and elongated in the rightward/leftward direction. The drive unit side gear cover 62 has a size (a vertical dimension and a front-to-rear dimension) substantially the same as that of the left side wall 56L of the developing cartridge 25. The drive unit side gear cover 62 covers at least a part of the gear portion 64.

(83) The drive unit side gear cover 62 is provided with a coupling collar 65, a first engagement portion 66, an operated portion 67, and a guided portion 68.

(84) The coupling collar 65 is formed in a generally cylindrical shape protruding leftward from a left side wall of the drive unit side gear cover 62 at a substantially front-to-rear center portion of the drive unit side gear cover 62. The coupling collar 65 has a right end portion in communication with an interior of the drive unit side gear cover 62.

(85) The first engagement portion 66 is a protrusion protruding rearward from a rear edge of the coupling collar 65 and also elongated in the rightward/leftward direction. The first engagement portion 66 is adapted to move the developing roller 16 away from the photosensitive drum 15 and to be abutable on the corresponding separating member 44. The first engagement portion 66 is positioned between the axis A2 of the coupling portion 60 and the axis A1 of the developing roller 16 such that a rear surface 91 of the first engagement portion 66 is overlapped with the developing roller 16 when projected in the rightward/leftward direction (FIG. 7). The rear surface 91 (FIG. 7) of the first engagement portion 66 is a plane extending in the vertical direction (in a direction from a lower front side to an upper rear side of the process cartridge 11 in FIG. 7). The rear surface 91 is perpendicular to a confronting direction in which the photosensitive drum 15 confronts the developing roller 16 (a direction from a lower rear side to an upper front side of the process cartridge 11 in FIG. 7).

(86) The operated portion 67 is integrally provided with a pressed portion 69 and a restricted portion 70.

(87) The pressed portion 69 is connected to a front edge of the coupling collar 65. The pressed portion 69 is a protrusion that protrudes leftward from a left surface of the drive unit side gear cover 62. The pressed portion 69 also extends in the frontward/rearward direction.

(88) The restricted portion 70 is connected to a front edge of the pressed portion 69. The restricted portion 70 is a protrusion that protrudes leftward from the left surface of the drive unit side gear

cover **62**. The restricted portion **70** extends diagonally frontward and upward from the front edge of the pressed portion **69**. That is, the restricted portion **70** is disposed opposite to the first engagement portion **66** with respect to the coupling portion **60**.

(89) The guided portion **68** is spaced apart from a front portion of the restricted portion **70** at a position downward of the restricted portion **70**. The guided portion **68** is formed in a generally wedge shape that protrudes leftward from the left surface of the drive unit side gear cover **62**. The guided portion **68** has a lower surface **77** sloping upward toward a left side thereof. The guided portion **68** is adapted to be abutable on the lock lever **38**.

(90) Further, the drive unit side gear cover **62** is fixed to the left side wall **56L** by screws such that a left end portion of the developing coupling **61** is fitted into the coupling collar **65**. Incidentally, the recessed connection portion **63** is exposed through a left end portion of the coupling collar **65**.

(91) (2-3) Detection Unit

(92) As shown in FIG. **4**, the detection unit **53** includes a detection gear **71** and a detection unit side gear cover **72**.

(93) The detection gear **71** is rotatably supported to the right side wall **56R**. The detection gear **71** is formed in a generally cylindrical shape extending in the rightward/leftward direction. The detection gear **71** is integrally provided with a detected end portion **73**.

(94) The detected end portion **73** is disposed at a right end portion of the detection gear **71**. The detected end portion **73** includes a pair of detected portions **74** and a connection portion **75**.

(95) Each detected portion **74** is disposed at each end portion of the detection gear **71** in a radial direction of the detection gear **71**. Each detected portion **74** is formed in a generally pillar shape extending in the rightward/leftward direction.

(96) The connection portion **75** has a generally flat plate shape and also has a generally rhombus shape in a side view. The connection portion **75** is connected to right edges of the detected portions **74**.

(97) A drive force from the developing coupling **61** is transmitted to the detection gear **71** via the gear train (not shown). When a new and unused developing cartridge **25** is mounted in the main casing **2**, the detection gear **71** rotates at a predetermined driving amount. At this time, each detected portion **74** is detected by a detection unit (not shown) provided at the main casing **2**. Based on this detection, information related to the developing cartridge **25** (new or used, and type) is determined on the main casing **2** side.

(98) The detection unit side gear cover **72** is formed in a generally square pillar shape whose right end is closed, and elongated in the rightward/leftward direction. The detection unit side gear cover **72** has a size (a vertical dimension and a front-to-rear dimension) sufficient to cover the detection gear **71**. That is, the detection unit side gear cover **72** covers at least a part of the detection gear **71**.

(99) Further, the detection unit side gear cover **72** is formed with a detection gear exposure opening **78**. Further, the detection unit side gear cover **72** is provided with a second engagement portion **76**.

(100) The detection gear exposure opening **78** has a generally circular shape in a side view. The detection gear exposure opening **78** penetrates a right side wall of the detection unit side gear cover **72** at a substantially front-to-rear center portion of the detection unit side gear cover **72** so that the detected end portion **73** of the detection gear **71** is exposed through the detection gear exposure opening **78**.

(101) The second engagement portion **76** is a protrusion protruding rearward from a rear edge of the detection unit side gear cover **72** and also elongated in the rightward/leftward direction. The second engagement portion **76** is adapted to move the developing roller **16** away from the photosensitive drum **15** and to be abutable on the corresponding separating member **44**. The second engagement portion **76** is positioned between the axis **A2** of the coupling portion **60** and the axis **A1** of the developing roller **16** such that a rear surface **92** of the second engagement portion **76** is overlapped with the developing roller **16** when projected in the rightward/leftward direction (FIG. **8**). The rear surface **92** (FIG. **8**) of the second engagement portion **76** is a plane extending in

the vertical direction (in the direction from the upper rear side to the lower front side of the process cartridge **11** in FIG. **8**), in the same manner as the rear surface **91** of the first engagement portion **66**. The rear surface **92** is perpendicular to the confronting direction in which the photosensitive drum **15** confronts the developing roller **16** (the direction from the lower rear side to the upper front side of the process cartridge **11** in FIG. **8**).

(102) Further, the detection unit side gear cover **72** is fixed to the right side wall **56R** by screws such that the detected end portion **73** of the detection gear **71** is exposed through the detection gear exposure opening **78**.

(103) 3. Attachment and Detachment of Developing Cartridge Relative to Drum Cartridge

(104) In order to mount the process cartridge **11** in the main casing **2**, initially, the developing cartridge **25** is attached to the drum cartridge **24**.

(105) To attach the developing cartridge **25** to the drum cartridge **24**, as shown in FIG. **5**, a rear end portion of the developing cartridge **25** is inserted into a rear end portion of the developing cartridge mounting portion **36**.

(106) Then, the developing roller **16** is brought into contact with the photosensitive drum **15** from a front side thereof. Further, the guided portion **68** confronts the restricting portion **43** of the lock lever **38** from a top side thereof.

(107) Next, a front end portion of the developing cartridge **25** is pushed into a front end portion of the developing cartridge mounting portion **36** so that the front end portion of the developing cartridge **25** is pivotally moved about the rear end portion of the developing cartridge **25** as indicated by an arrow B in FIG. **5** in a clockwise direction as viewed from a left side.

(108) Then, as shown in FIG. **6A**, the lower surface **77** of the guided portion **68** of the developing cartridge **25** is brought into abutment with the right edge of the restricting portion **43** of the lock lever **38** from a top side thereof.

(109) When the front end portion of the developing cartridge **25** is further pushed into the front end portion of the developing cartridge mounting portion **36**, the developing cartridge **25** is pushed rightward while guided by the sloped lower surface **77** of the guided portion **68**, and the front end portion of the developing cartridge **25** is inserted into the front end portion of the developing cartridge mounting portion **36**.

(110) At the same time, the lock lever **38** is pressed frontward by the guided portion **68** of the developing cartridge **25**, and pivotally moved in the clockwise direction as viewed from a left side against the urging force from the urging member (not shown).

(111) Further, when the front end portion of the developing cartridge **25** is still further pushed into the front end portion of the developing cartridge mounting portion **36**, as shown in FIG. **6B**, the guided portion **68** is moved past the lock lever **38** at a right front side thereof. Hence, the developing cartridge **25** is completely mounted in the developing cartridge mounting portion **36**.

(112) Upon completion of mounting of the developing cartridge **25** in the developing cartridge mounting portion **36**, abutment of the guided portion **68** with the lock lever **38** is released.

(113) As a result, the lock lever **38** is pivotally moved in the counterclockwise direction as viewed from a left side by the urging force from the urging member (not shown). Hence, the lock lever **38** is again positioned at the lock position.

(114) At this time, the restricting portion **43** of the lock lever **38** confronts the restricted portion **70** of the developing cartridge **25** from a top side thereof. When the developing cartridge **25** is pivotally moved in the counterclockwise direction as viewed from a left side, the restricting portion **43** of the lock lever **38** is brought into engagement with the restricted portion **70** of the developing cartridge **25**. Hence, the restricting portion **43** of the lock lever **38** restricts further pivotal movement of the developing cartridge **25** in the counterclockwise direction as viewed from a left side. The lock lever **38** thus prohibits detachment of the developing cartridge **25** relative to the drum cartridge **24**.

(115) Further, when the lock lever **38** is positioned at the lock position, the lift portion **42** of the

lock lever **38** confronts the pressed portion **69** of the developing cartridge **25** from a bottom side thereof.

(116) Further, when the developing cartridge **25** has been completely mounted in the developing cartridge mounting portion **36**, as shown in FIG. 7, the pressure members **49** are in abutment with the front wall **57** of the cartridge frame **51** of the developing cartridge **25** from a front side thereof (an upper front side in FIG. 7). The developing cartridge **25** is therefore normally pressed rearward (diagonally below and rearward in FIG. 7) by the urging force from the urging members **50** that respectively urge the pressure members **49** while the developing cartridge **25** is accommodated in the developing cartridge mounting portion **36**.

(117) Further, the first engagement portion **66** of the drive unit side gear cover **62** confronts the corresponding separating member **44** from a front side thereof (an upper front side in FIG. 7), and the second engagement portion **76** of the detection unit side gear cover **72** confronts the corresponding separating member **44** from a front side thereof (an upper front side in FIG. 8).

(118) Incidentally, the operated portion **67** (the pressed portion **69** and the restricted portion **70**) and the guided portion **68** may be provided at the detection unit side gear cover **72**, and the lock lever **38** may be pivotally movably supported to the right side wall **32R**.

(119) Next, an operation for detaching the developing cartridge **25** from the drum cartridge **24** will be described. To detach the developing cartridge **25** from the drum cartridge **24**, initially, the operation portion **41** of the lock lever **38** is pressed, thereby pivotally moving the lock lever **38** in the clockwise direction as viewed from a left side against the urging force from the urging member (not shown). The lock lever **38** is thus positioned at the unlock position (not shown).

(120) Then, the restricting portion **43** of the lock lever **38** is retracted frontward from a top side of the restricted portion **70** of the developing cartridge **25**. At this time, the pressed portion **69** of the developing cartridge **25** is pressed upward by the lift portion **42** of the lock lever **38** to permit detachment of the developing cartridge **25** from developing cartridge mounting portion **36**.

(121) As a result, the front end portion of the developing cartridge **25** is lifted upward, thereby moving the developing cartridge **25** away from the developing cartridge mounting portion **36** of the drum cartridge **24**.

(122) Then, the user holds the front end portion of the developing cartridge **25** to move the developing cartridge **25** upward, thereby detaching the developing cartridge **25** from the developing cartridge mounting portion **36** of the drum cartridge **24**.

(123) Detachment of the developing cartridge **25** from the drum cartridge **24** is thus completed.

(124) To mount the process cartridge **11** in the main casing **2**, as shown in FIG. 1, the top cover **6** is opened. The process cartridge **11** is positioned at a predetermined position within the main casing **2**, and mounted in the main casing **2** from a top side thereof such that the rear portion of the process cartridge **11** is disposed at a lower rear side of the printer **1** and the front portion of the process cartridge **11** is disposed at an upper front side of the printer **1**.

(125) Mounting of the process cartridge **11** in the main casing **2** is thus completed.

(126) 4. Separation of Developing Cartridge from Photosensitive Drum

(127) When a color image is printed in the printer **1**, the developing roller **16** of each developing cartridge **25** is in contact with the corresponding photosensitive drum **15** (FIG. 1).

(128) However, when a monochromatic image (black color image) is printed, the developing roller **16** is in contact with the corresponding photosensitive drum **15** in the black process cartridge **11K** while each developing roller **16** is moved away from the corresponding photosensitive drum **15** in the color process cartridges **11** (i.e. the yellow process cartridge **11Y**, the magenta process cartridge **11M**, and the cyan process cartridge **11C**). That is, each developing roller **16** is movable away from the corresponding photosensitive drum **15** frontward (frontward and upward in FIGS. 7 and 8). The direction in which the developing roller **16** is moved away from the corresponding photosensitive drum **15** corresponds to a separating direction and a perpendicular direction perpendicular to the axial direction.

(129) To separate the developing roller **16** from the photosensitive drum **15**, as shown in FIGS. 7, 8, and 9, a pair of levers **81** is provided at the main casing **2** (indicated by broken lines in FIGS. 7, 8, and 9).

(130) Each lever **81** is adapted to push the contacted portion **46** of each separating member **44** rearward (downward and rearward in FIGS. 7 and 8), which causes the separating member **44** to pivotally move in the clockwise direction as viewed from a left side. The pressing portion **47** of the left separating member **44** is therefore brought into abutment with the first engagement portion **66** of the drive unit side gear cover **62** from a rear side thereof (a lower rear side in FIG. 7), and also the pressing portion **47** of the right separating member **44** is brought into abutment with the second engagement portion **76** of the detection unit side gear cover **72** from a rear side thereof (a lower rear side in FIG. 8).

(131) As both of the separating members **44** are further pivotally moved by the levers **81** in the clockwise direction as viewed from a left side, the pressing portion **47** of the left separating member **44** presses the rear surface **91** of the first engagement portion **66** forward (forward and upward in FIG. 9, separating direction), and the pressing portion **47** of the right separating member **44** presses the rear surface **92** of the second engagement portion **76** forward.

(132) The developing cartridge **25** is thus moved forward (forward and upward in FIG. 9, separating direction) against the urging force from the pressure members **49**, as shown in FIG. 9.

(133) As a result, the developing roller **16** is moved away from the photosensitive drum **15**.

(134) 5. Operations and Effects

(135) (1) According to the above-described process cartridge **11**, as shown in FIGS. 3 and 4, the drive unit side gear cover **62** of the developing cartridge **25** includes the first engagement portion **66**, and the detection unit side gear cover **72** includes the second engagement portion **76**. Each of the first engagement portion **66** and the second engagement portion **76** is engageable with the separating member **44** of the drum cartridge **24**.

(136) Therefore, it is not necessary to provide an engagement portion at the cartridge frame **51**, avoiding the drive unit side gear cover **62** and the detection unit side gear cover **72**. Accordingly, the first engagement portion **66** and the second engagement portion **76** are positionally efficiently arranged. Thus, a compact developing cartridge **25** can be provided.

(137) Further, while the first engagement portion **66** can be provided at the drive unit side gear cover **62** and the second engagement portion **76** can be provided at the detection unit side gear cover **72**. The degree of freedom in design of the developing cartridge **25** can therefore be increased.

(138) (2) Further, according to the above-described process cartridge **11**, as shown in FIGS. 7 and 8, the first engagement portion **66** has the rear surface **91** extending in a direction perpendicular to the confronting direction in which the developing roller **16** and the photosensitive drum **15** confront each other (the direction from the upper front side to the lower rear side of the process cartridge **11**). Likewise, the second engagement portion **76** has the rear surface **92** extending in a direction perpendicular to the confronting direction.

(139) With this configuration, the rear surface **91** of the first engagement portion **66** and the rear surface **92** of the second engagement portion **76** can be stably pressed diagonally upward and frontward (in the direction in which the developing roller **16** is separated from the photosensitive drum **15**). Hence, the developing roller **16** can be stably moved away from the photosensitive drum **15**.

(140) (3) Further, according to the above-described process cartridge **11**, as shown in FIGS. 7 and 8, each of the rear surface **91** of the first engagement portion **66** and the rear surface **92** of the second engagement portion **76** is overlapped with the developing roller **16** when projected in the rightward/leftward direction.

(141) With this configuration, the developing roller **16** can be separated from the photosensitive drum **15** by pressing the rear surface **91** of the first engagement portion **66** and the rear surface **92**

of the second engagement portion **76** at a position adjacent to the developing roller **16**.

(142) Therefore, separation of the developing roller **16** from the photosensitive drum **15** can be reliably achieved.

(143) (4) Further, according to the above-described process cartridge **11**, as shown in FIG. 5, the drive unit side gear cover **62** includes the restricted portion **70** abutable on the restricting portion **43** of the lock lever **38** of the drum cartridge **24**.

(144) With this configuration, while the degree of freedom in designing the developing cartridge **25** can be increased, detachment of the developing cartridge **25** from the drum cartridge **24** can be restricted.

(145) (5) Further, according to the above-described process cartridge **11**, as shown in FIGS. 7 and 8, the restricted portion **70** is positioned frontward of the developing coupling **61** when projected in the rightward/leftward direction.

(146) With this configuration, detachment of the developing cartridge **25** from the drum cartridge **24** can be restricted at a position frontward of the developing roller **16** and relatively away from the developing roller **16**.

(147) (6) Further, according to the above-described process cartridge **11**, as shown in FIG. 5, the drive unit side gear cover **62** includes the pressed portion **69** that is pressed by the lift portion **42** of the lock lever **38** of the drum cartridge **24**.

(148) With this configuration, while the degree of freedom in design of the developing cartridge **25** can be increased, the developing cartridge **25** can be lifted up from the drum cartridge **24** using the drive unit side gear cover **62**.

(149) (7) Further, according to the above-described process cartridge **11**, as shown in FIG. 5, the pressed portion **69** and the restricted portion **70** are connected to each other. This configuration can easily provide strength with the pressed portion **69** and the restricted portion **70**.

(150) (8) Further, according to the above-described process cartridge **11**, as shown in FIG. 5, the drive unit side gear cover **62** includes the guided portion **68**. The guided portion **68** is abutable on the lock lever **38** of the drum cartridge **24**, thereby pressing the guided portion **68** rightward.

(151) In conjunction with abutment of the guided portion **68** of the drive unit side gear cover **62** with the lock lever **38**, the developing cartridge **25** can be pressed rightward. Since the right edge of the developing cartridge **25** aligns against the right side wall **32R** of the drum cartridge **24**, the developing cartridge **25** can be subjected to positioning relative to the drum cartridge **24**.

(152) (9) Further, according to the above-described process cartridge **11**, as shown in FIGS. 7 and 8, the drum cartridge **24** includes the pair of separating members **44** for pressing the first engagement portion **66** and the second engagement portion **76** frontward and upward (in the direction in which the developing roller **16** is moved away from the photosensitive drum **15**).

(153) Therefore, separation of the developing cartridge **25** from the photosensitive drum **15** can be reliably achieved.

6. Second Embodiment

(154) A process cartridge **111** including a drum cartridge **124** and a developing cartridge **125** according to a second embodiment of the present invention will be described while referring to FIGS. 10 and 11.

(155) In the following description, parts and components appearing in the second embodiment and the same as those in the first embodiment will be designated by the same reference numerals as those in the first embodiment to avoid duplicating description, and only parts and components differing from those of the first embodiment will be described.

(156) In the above-described first embodiment, each pressure member **49** is brought into abutment with the front wall **57** of the cartridge frame **51** of the developing cartridge **25** from a front side thereof, thereby pressing the developing cartridge **25** rearward. However, in the second embodiment, as shown in FIGS. 10 and 11, the drum cartridge **124** is provided with two pressure members **149**, one adapted to press a front edge of the drive unit side gear cover **62** and another

adapted to press a front edge of the detection unit side gear cover **72**. More specifically, one of the pressure members **149** is brought into abutment with the front edge of the drive unit side gear cover **62** from a front side thereof to press the developing cartridge **25** rearward. Likewise, remaining one of the pressure members **149** is brought into abutment with the front edge of the detection unit side gear cover **72** from a front side thereof to press the developing cartridge **25** rearward.

(157) A front wall **133** of the drum cartridge **124** is provided with two pressure member retaining portions **48**. One of the pressure members **149** is retained in one of the pressure member retaining portions **148** provided at a left end portion of the front wall **133** so as to confront the drive unit side gear cover **62** from a front side thereof. Likewise, remaining one of the pressure members **149** is retained in remaining one of the pressure member retaining portions **148** provided at a right end portion of the front wall **133** so as to confront the detection unit side gear cover **72** from a front side thereof. Further, a drive unit side gear cover **162** of the drive unit **52** is formed in a generally square pillar shape, and has a lower front edge formed in a generally arcuate shape in a side view.

(158) Further, a detection unit side gear cover **172** of the detection unit **53** is formed in a generally square pillar shape, and has a lower front edge formed in a generally arcuate shape in a side view. More specifically, the detection unit side gear cover **172** has a front portion having a shape substantially the same as that of the cartridge frame **51** when projected in the rightward/leftward direction. The detection unit side gear cover **172** has such a size (a front-to-rear dimension and a vertical dimension) that the front portion of the cartridge frame **51** can be overlapped with (cover) the front portion of the detection unit side gear cover **172** when projected in the rightward/leftward direction.

(159) In such a state that the developing cartridge **125** is attached to the drum cartridge **124**, one of the pressure members **149** is abutable on the lower front edge of the drive unit side gear cover **162** (a portion having an arcuate shape in a side view) from a front side thereof, and remaining one of the pressure members **149** is abutable on the lower front edge of the detection unit side gear cover **172** (a portion having an arcuate shape in a side view) from a front side thereof. The developing cartridge **125** is therefore normally pressed rearward by an urging force of the urging members **50** that urges the respective pressure members **149**.

(160) In other words, the lower front edge of the drive unit side gear cover **162** (the portion having an arcuate shape in a side view) and the lower front edge of the detection unit side gear cover **172** (the portion having an arcuate shape in a side view) are pressed by the respective pressure members **149** in a direction in which the developing roller **16** contacts the photosensitive drum **15**.

(161) According to the second embodiment, the developing roller **16** is pressed toward the photosensitive drum **15** using the drive unit side gear cover **162** and the detection unit side gear cover **172**.

(162) In the second embodiment, operations and effects similar to those of the first embodiment can also be obtained.

7. Third Embodiment

(163) A process cartridge **211** including a drum cartridge **224** and a developing cartridge **225** according to a third embodiment of the present invention will be described while referring to FIGS. **12** through **16**.

(164) In the following description, parts and components appearing in the third embodiment and the same as those in the first embodiment will be designated by the same reference numerals as those in the first embodiment to avoid duplicating description, and only parts and components differing from those of the first embodiment will be described.

(165) In the third embodiment, as shown in FIG. **12**, the developing cartridge **225** is provided with a drive unit side gear cover **262** and a detection unit side gear cover **272**. The drive unit side gear cover **262** has an attachment operation guide **96** and three detachment operation guides **97**. The attachment operation guide **96** is adapted to guide a user's attachment operation for attaching the developing cartridge **225** to the drum cartridge **224**. The detachment operation guides **97** are

adapted to guide a user's detachment operation for detaching the developing cartridge **225** from the drum cartridge **224**. The attachment operation guide **96** and the detachment operation guide **97** may be provided at the detection unit side gear cover **272**.

(166) The attachment operation guide **96** is generally flat plate-like shaped, having a generally triangle shape in a side view. The attachment operation guide **96** protrudes frontward from a front surface of the drive unit side gear cover **262** at a right end portion of the drive unit side gear cover **262**. The attachment operation guide **96** has a bottom surface **98**, extending linearly in the frontward/rearward direction in a side view.

(167) Further, the drum cartridge **24** has a front wall **233** whose rear end wall is formed with a guide accommodating opening **99**. The guide accommodating opening **99** penetrates the rear end wall. The guide accommodating opening **99** is adapted to accommodate the attachment operation guide **96** therein.

(168) The guide accommodating opening **99** has a generally rectangular shape elongated in the vertical direction along a movement path of the attachment operation guide **96** (indicated by an imaginary line C in each of FIGS. **14** and **15**) during the attachment operation of the developing cartridge **225** to the drum cartridge **224**.

(169) Three detachment operation guides **97** are arranged juxtaposed with and spaced apart from each other in the frontward/rearward direction (specifically, in FIG. **12**, a direction from an upper front side to a lower rear side of the developing cartridge **225**). Each detachment operation guide **97** is generally flat plate-like shaped, having a generally triangle shape in a front view, protruding leftward from a left surface of an upper front end portion of the drive unit side gear cover **262**.

(170) To attach the developing cartridge **225** to the drum cartridge **224**, in the same manner as the first embodiment described above, the user initially inserts the rear end portion of the developing cartridge **225** into the rear end portion of the developing cartridge mounting portion **36**.

(171) Here, as shown in FIG. **13**, if the rear end portion of the developing cartridge **225** is not sufficiently inserted into the developing cartridge mounting portion **36**, the attachment operation guide **96** is brought into abutment with the front wall **33** of the drum cartridge **224** from a top side thereof.

(172) Abutment of the attachment operation guide **96** with the front wall **33** restricts further downward movement of the front end portion of the developing cartridge **225** into the developing cartridge mounting portion **36**.

(173) Thus, the user has to stop further attachment operation of the developing cartridge **225** to the drum cartridge **224**. As a result, the user can realize that insertion of the rear end portion of the developing cartridge **225** into the developing cartridge mounting portion **36** is insufficient.

(174) Then, when the user adequately inserts the rear end portion of the developing cartridge **225** into the developing cartridge mounting portion **36**, the developing roller **16** is brought into contact the photosensitive drum **15** from a front side thereof. Further, as shown in FIG. **14**, the attachment operation guide **96** is positioned rearward of the front wall **33** of the drum cartridge **224**.

(175) Next, in the same manner as the first embodiment described above, the front end portion of the developing cartridge **225** is pushed into the front end portion of the developing cartridge mounting portion **36** so that the front end portion of the developing cartridge **225** is pivotally moved in the clockwise direction as viewed from a left side about the rear end portion of the developing cartridge **225**.

(176) Then, as described above, and as shown in FIG. **15**, the developing cartridge **225** is completely mounted in the developing cartridge mounting portion **36**. At this time, the front end portion of the attachment operation guide **96** is accommodated within the guide accommodating opening **99**.

(177) Attachment of the developing cartridge **225** to the drum cartridge **224** is thus completed.

(178) Further, to detach the developing cartridge **225** from the drum cartridge **224**, in the same manner as the first embodiment described above, the user presses the operation portion **41** of the

lock lever **38** against the urging force from the urging member (not shown) of the lock lever **38** to pivotally move the lock lever **38** in the clockwise direction as viewed from a left side. The lock lever **38** is thus positioned at the unlock position (not shown).

(179) At this time, as shown in FIG. **16**, as the user moves his/her fingers **100** downward along the detachment operation guides **97** from a top side thereof, the user's fingers **100** are guided to the operation portion **41** of the thickness-regulating blade **28** along sloped surfaces of the detachment operation guides **97**.

(180) As described above, the restricting portion **43** of the lock lever **38** is retracted frontward from a top side of the restricted portion **70** of the developing cartridge **225**. At this time, the lift portion **42** of the lock lever **38** presses the pressed portion **69** of the developing cartridge **225** upward, thereby lifting the front end portion of the developing cartridge **225** upward, and moving the developing cartridge **225** away from the developing cartridge mounting portion **36** of the drum cartridge **224**.

(181) The user then holds the front end portion of the developing cartridge **225** to move the developing cartridge **225** upward, thereby detaching the developing cartridge **225** from the developing cartridge mounting portion **36** of the drum cartridge **224**.

(182) Detachment of the developing cartridge **225** relative to the drum cartridge **24** is thus completed.

(183) According to the third embodiment, when the user operates the operation portion **41**, the user's fingers **100** can be guided to the operation portion **41**, using the drive unit side gear cover **62**.

(184) In the third embodiment, operations and effects similar to those of the first embodiment can also be obtained.

(185) While the present invention has been described in detail with reference to the present embodiments thereof, it would be apparent to those skilled in the art that various changes and modifications may be made therein without departing from the spirit of the present invention.

Claims

1. A developing cartridge comprising: a developing roller extending in a first direction, the developing roller being rotatable about a developing axis extending in the first direction; a frame configured to accommodate developing agent therein, the frame including one end portion and another end portion separated from the one end portion of the frame in a second direction, the developing roller being positioned at the one end portion of the frame; a coupling rotatable about a coupling axis extending in the first direction, the coupling being positioned at one side of the frame in the first direction; a cover being positioned at another side of the frame, the cover having a first distal end portion and a second distal end portion spaced apart from the first distal end portion of the cover in the second direction, the first distal end portion of the cover being closer to the developing roller than the second distal end portion of the cover is to the developing roller in the second direction; and a first engagement protrusion extending in the first direction, the first engagement protrusion being positioned at the another side of the frame, the first engagement protrusion being positioned at the first distal end portion of the cover; wherein a distal end of the first engagement protrusion is positioned between the developing axis and the coupling axis in the second direction.

2. The developing cartridge according to claim 1, wherein the first engagement protrusion is positioned on the first distal end portion of the cover.

3. The developing cartridge according to claim 1, wherein the first engagement protrusion extends from the first distal end portion of the cover.

4. The developing cartridge according to claim 1, wherein the first engagement protrusion is integrally formed with the first distal end portion of the cover.

5. The developing cartridge according to claim 1, further comprising: a coupling cover including a

coupling collar extending in the first direction, the coupling collar covering a portion of the coupling, the coupling collar having an opening configured to allow a portion of the coupling to be exposed.

6. The developing cartridge according to claim 1, further comprising: a second engagement protrusion extending in the first direction, the second engagement protrusion being positioned at the one side of the frame.

7. The developing cartridge according to claim 1, wherein the developing roller includes a developing roller shaft extending in the first direction.

8. The developing cartridge according to claim 7, wherein the first engagement protrusion is positioned between the developing roller shaft and the first distal end portion of the cover in the second direction.

9. The developing cartridge according to claim 1, wherein the first engagement protrusion protrudes toward the developing roller.

10. The developing cartridge according to claim 1, wherein the first engagement protrusion is abutable on a separating member of a drum cartridge in a state where the developing cartridge is received in the drum cartridge.

11. The developing cartridge according to claim 1, further comprising: a detection protrusion movable relative to the frame according to rotation of the coupling, the detection protrusion being positioned at the another side of the frame.
